

Human hybrid electric

Human-powered transport means to make use of human muscle power for the transport of person(s) and/or goods. This was the only system of transportation before the invention of the steam engine by James Watt, which then contributed substantially to the industrial revolution. With time, motorisation has increased speed and load capacity, but still human-powered transport in many of its forms remains popular due to several reasons such as cost and convenience.

While showering praise on EVs, we ignored the effects of ever-increasing pollution that is being produced during the generation of electric energy, an essential requirement to charge batteries of EVs as well as in the manufacturing of batteries. In order to minimise this impact, engineers are thinking to imbibe human power in electric vehicles by developing human hybrid electric vehicles.

A human hybrid electric vehicle is a system that makes use of human power in an electric vehicle called a hybrid human-powered vehicle, whose drivetrain consists of a person, an electric motor/generator, and one or more electricity storage device(s). Many developments have already taken place towards the development of human hybrid electric vehicles, and time is not far when we will have such transport vehicles all around us.

Developmental status

1. TWIKE (a Tesla company project) is the world's oldest only human electric hybrid car. This car was developed and produced initially in Switzerland, but then production moved to Germany after the initial investors became insolvent in 2002, and a group of German importers joined together to purchase the equipment and continue with TWIKE production under the company named FINE Mobile.

TWIKE is an ambitious concept today as it was back when it first became a rare oddity in the world of transportation. Presently, instead of a standard steering wheel, navigation of the car is done through a joystick. Under the driver's feet is the direct drive pedal power system, which



Minnesota-based company has built a three-wheeled hybrid electric vehicle
(Credit: <https://inhabitat.com/wp-content>)

transfers human power input straight to the drivetrain. The system also captures energy from regenerative braking. At higher speeds, the vehicle is powered mostly by onboard lithium-ion batteries. The top speed reported to be achieved with the car at present is around 89 to 104 kilometres per hour.

Similar projects are continuously being reported by various companies. For instance, a Minnesota-based company has built a three-wheeled hybrid electric vehicle weighing about 317kg that gets 121km per charge, and the vehicle is street legal and considered a motorcycle. Its carbon fibre chassis looks like a kayak and is reinforced with structural foam, similar to an airplane. The vehicle uses a regenerative brake system, which feeds the electricity back into the batteries when the brakes are applied. It also has a solar panel mounted on the roof.

2. An electric motorbike with pedals, eRokit (developed and produced in Hennigsdorf in Brandenburg, Berlin), is described as a 'human hybrid' as it's powered by a combination of human and electric drives. The eRokit puts the rider back into the centre of mobility of moving, depending on the rider's wish. Further, the use of a smart pedal concept gives the rider an opportunity to act with a human hybrid bike more conveniently and comfortably.

The bike's clutch-free and intuitive

pedal power system makes eRokit safe to drive. It is an emission-free ride with a hundred per cent emotion.

The bike can reach a speed of close to 55.9mph, and its battery range is over 120 kilometres. With this new technology of human hybrid electric vehicle, a person can travel between 5km and 45km every day together with fun riding. About thirty prototypes have already been tested for more than 100,000 kilometres in total, and the market version will be available soon.

Conclusion

In a world that is wholly dependent on transportation on one hand and fossil fuels on the other, harvesting human kinetic energy along with the future of efficient, light, cost-effective electric vehicles will provide an era of environment-friendly and healthy transportation mode—human hybrid electric vehicles. By using a pedal-assisted drivetrain system, we will be able to reduce the consumption rate of battery power by the motor, which will increase the battery life.

The reason why many people are inclined towards conventional vehicles is that they can reach places without having to do more than pressing on the gas pedal. In short, developing people-friendly human hybrid electric vehicles will definitely prove to be an attractive mode of transportation in the future. **EFY**